



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
24-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

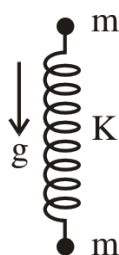
1. A mass m is hung on an ideal massless spring. Another equal mass is connected to the other end of the spring. The whole system is at rest. At $t = 0$, m is released and the system falls freely under gravity. Assume that natural length of the spring is L_0 , its initial stretched length is L and the acceleration due to gravity is g . What is distance between masses as function of time.

(A) $L_0 + (L - L_0) \cos \sqrt{\frac{2K}{m}} t$

(B) $L_0 + (L - L_0) \cos \sqrt{\frac{K}{m}} t$

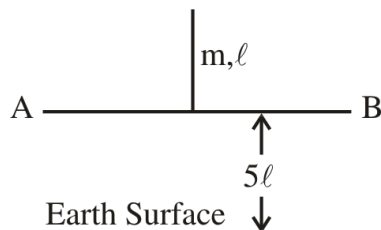
(C) $L_0 - 2(L + 2L_0) \cos \sqrt{\frac{2K}{m}} t$

(D) $L_0 + (L - L_0) \sin \sqrt{\frac{2K}{m}} t$



Space for Rough Work

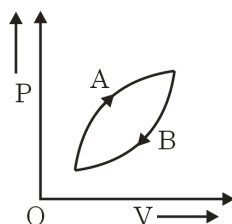
2. Consider a situation in which a uniform gravitation field is present below line AB and no gravity above line AB. Then the speed with which a rod shown in the figure will strike earth surface when released from the position shown is



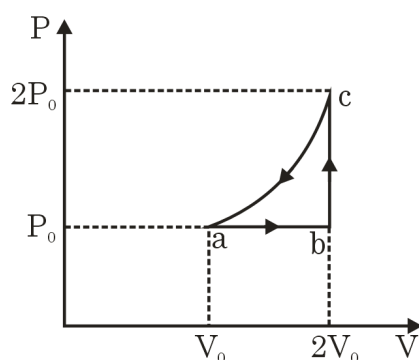
- (A) $\sqrt{gl}$ (B) $2\sqrt{gl}$ (C) $3\sqrt{gl}$ (D) $4\sqrt{gl}$
3. A non-conducting solid sphere of radius 30cm and relative permittivity 1 has the volume charge density $\rho = \left(\frac{5}{2\pi} \mu\text{C}/\text{m}^3 \right) \left(1 - \frac{r}{30\text{cm}} \right)$ where r (in cm) is the distance of any point in the space from the centre of the sphere. Then choose the correct statement(s).
- (A) Density of electric field lines is maximum at $r = 25$ cm.
- (B) Density of electric field lines is maximum at $r = 20$ cm.
- (C) Electrical energy stored in the space $r = 30$ cm to $r \rightarrow \infty$ is same as that of a solid non-conducting sphere carrying charges of $0.0225 \mu\text{C}$ and radii 30 cm having uniform volume charge density in the same space $r = 30$ cm to $r \rightarrow \infty$.
- (D) Variation of magnitude of electric field due to the sphere with respect to radial distance r is rectangular hyperbola in the region $r = 30$ cm to $r \rightarrow \infty$.
4. A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A, contained in the vessel is 2 times that of B :-
- (A) Pressure exerted by gas B is eight times of that exerted by gas A.
- (B) Number of molecules of gas B in the cylinder is eight times that of gas A.
- (C) Average kinetic energy per molecule of gas A is equal to that of gas B.
- (D) Root mean square value of translational speed of gas B is four times that of A.

Space for Rough Work

5. In the cyclic process shown in the given figure, ΔU_1 and ΔU_2 represent the change in internal energy in process A and B, respectively. If ΔQ be the net heat given to the system in the process and ΔW be the work done by the system in the process, then



- (A) $\Delta U_1 + \Delta U_2 = 0$ (B) $\Delta U_1 - \Delta U_2 = 0$ (C) $\Delta Q - \Delta W = 0$ (D) $\Delta Q + \Delta W = 0$
6. One mole of an ideal monatomic gas has initial temperature T_0 , is made to go through the cycle abca shown in the given figure. If U denotes the internal energy, then choose the correct alternative.



- (A) $U_c > U_b > U_a$ (B) $U_c - U_b = 3RT_0$
 (C) $U_c - U_a = \frac{9RT_0}{2}$ (D) $U_b - U_a = \frac{3RT_0}{2}$

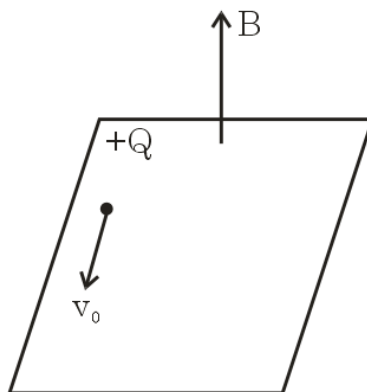
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

On a rough horizontal surface (coefficient of friction = μ) a charged particle of charge $+Q$ and mass m is projected with an initial velocity v_0 . The charge moves on the surface and eventually comes to rest. Magnetic field B is uniform and perpendicular to the surface as shown.



7. Which of the following statement(s) about the velocity of the particle is/are correct :- (as long as it is non zero)
- (A) The magnitude at any time is $v_0 - \mu gt$.
 - (B) The force of magnetic field is perpendicular to it.
 - (C) The velocity vector rotates at a constant angular velocity.
 - (D) The velocity vector rotates at a variable angular velocity.

Space for Rough Work

8. Choose the correct statement (s) about the path followed by the particle.

(A) The path is circular.

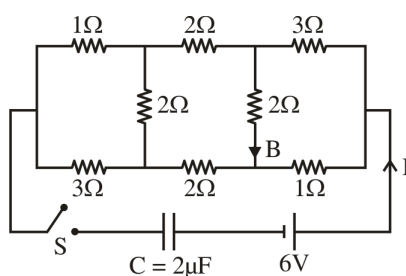
(B) The length of the path is $\frac{v_0^2}{2\mu g}$.

(C) The particle stops at a distance $\frac{3v_0^2}{2\mu g}$ from the initial position.

(D) The particle stops at a time $\frac{v_0}{\mu g}$ after being projected.

Paragraph for Questions 9 and 10

Consider the circuit shown in figure. Battery is ideal and capacitor was initially uncharged. If at $t = 0$ switch is closed, then answer the following question :



9. Current in the direction B shown in diagram at $t = 0$ (Just after the switch is closed) will be :-

- (A) $\frac{3}{8}$ A (B) $\left(\frac{-3}{8}\right)$ A (C) $\left(\frac{4}{5}\right)$ A (D) $\left(\frac{-4}{5}\right)$ A

10. Current I as a function of t (in μ s) will be :-

- (A) $I = \frac{9}{4}e^{-3t/8}$ (B) $I = \frac{3}{2}e^{-3t/10}$ (C) $I = \frac{9}{4}e^{-3t/16}$ (D) $I = \frac{3}{2}e^{-3t/5}$

Space for Rough Work

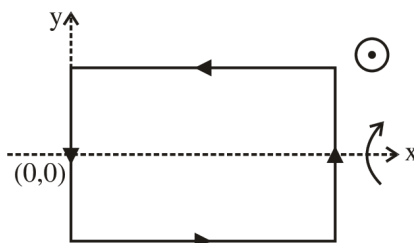
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

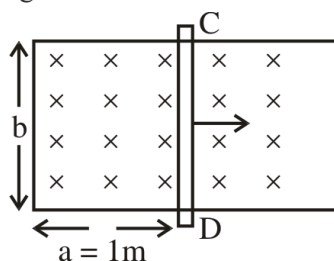
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A square conducting loop of side length $a = 4\text{m}$ carrying current $i_0 = 3\text{A}$ (anticlockwise) is placed in x - y plane. where magnetic field varies as $B = B_0 \left(1 + \frac{x}{4}\right) \hat{k}$ where $B_0 = \frac{1}{8}\text{T}$. Wire frame is so fixed that it can rotate freely about the central line as shown in figure work done by external agent in joule in rotating this loop slowly by 90° .

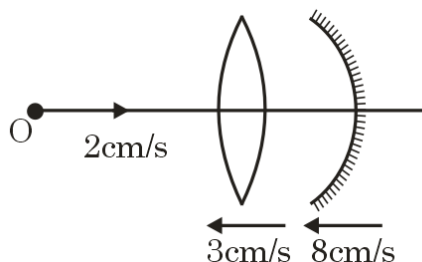


2. A 'U' shaped conducting frame is fixed in space. A conducting rod CD lies at rest on the smooth frame. The frame is kept in magnetic field B , which is perpendicular to the plane of frame. At time $t = 0$, the magnitude of magnetic field begins to change with time t as $B = \frac{B_0}{1 + 5t}$, where B is in tesla, t is in seconds and B_0 is positive constant for no current to be even induced in frame the constant speed (in m/s) with which the rod should moved starting from time $t = 0$ is

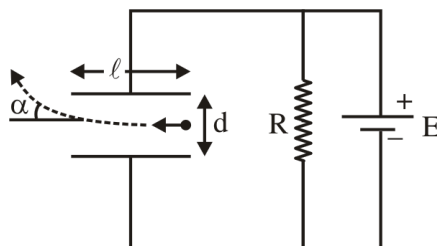


Space for Rough Work

3. An equiconvex lens of refractive index $\frac{3}{2}$ and of radius of curvature 16 cm is placed co-axially at a distance of 90 cm in front of concave mirror of radius of curvature 60 cm. The surrounding medium is air. A point object O is placed on the common axis of the lens and the mirror at a distance 90 cm left of the lens. At an instant ($t = 0$) the object, the lens and the mirror start moving with a speed 2 cm/s, 3 cm/s and 8 cm/s respectively in the direction shown in the figure. If the time after which the object and its final image coincide for the second time is $\frac{58}{N}$ second then N.

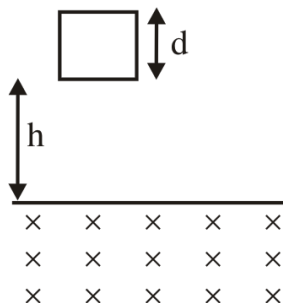


4. A parallel plate capacitor with plate of length ℓ is included in a circuit as shown in fig. Given are, e.m.f. of the current source, its internal resistance r and the distance d between the plates. As electron with a velocity u flies into the capacitor, so that the electron flies out of the capacitor at an angle 37° to the plates. Find the value of R in ohm (assuming circuit in steady state and neglect relativistic effect). Take $\ell = 91$ cm, $E = 3$ V, $r = 2\Omega$, $d = \frac{1}{3}$ mm, $u = 4 \times 10^7$ m/s, $M_e = 9.1 \times 10^{-31}$ kg and $e = 1.6 \times 10^{-19}$ C.



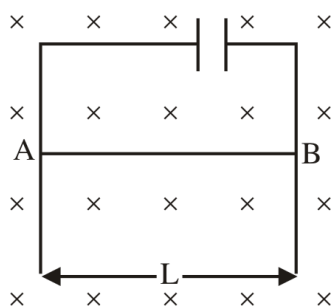
Space for Rough Work

5. A small conducting square loop of side length $d = 10 \text{ cm}$, is at height h above the boundary of a region of uniform magnetic field $B_0 = 1 \text{ T}$ (into the plane of the paper as shown in figure). The mass of the square loop is $m = 100 \text{ gram}$ and resistance $R = 0.1 \Omega$. The loop is dropped from such position and after some time it enters the magnetic field region. Find the height ' h ' in metre so that velocity of the loop will remain constant during the time interval the loop will be entering into the magnetic field region ($g = 10 \text{ m/s}^2$)



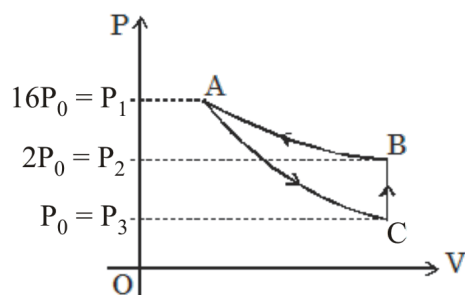
6. Two metals bars are fixed vertically and are connected on the top by a capacitor C . A sliding conductor AB of length L slides without friction with its ends in contact with the bars. The arrangement is placed in a uniform horizontal magnetic field directed normal to the plane of the figure. The conductor is released from rest. Find the displacement (x) in meter of the conductor at time $t = 2 \text{ sec}$.

[Given, $M = 100 \text{ gm}$, $g = 10 \text{ m/s}^2$, $B = 100 \text{ Tesla}$, $L = 1 \text{ m}$ and $C = 10 \mu\text{F}$]

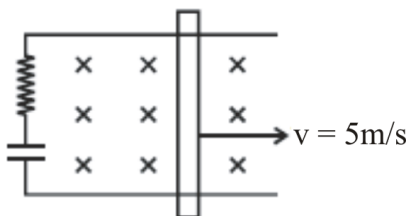


Space for Rough Work

7. In the figure three process on a gas are shown. The value of C_V is αR for the gas. If AC represents adiabatic process and BA represents isothermal process then α ?



8. A resistanceless rod of length 3m is being pulled in a uniform vertical magnetic field of magnitude 0.8 T with a speed of 5 m/s. The resistance of circuit is 4Ω and capacitance is $6\mu\text{F}$. Initially there is no charge on the capacitor. What is the current (in A) in circuit when charge on capacitor becomes $24\mu\text{C}$?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following statement is/are incorrect :

- (A) Dipositive zinc exhibits paramagnetism due to loss of two electrons from 3d-orbital of neutral atom.
- (B) The magnetic moment of an atom is related to the number of unpaired electrons in its electronic configuration.
- (C) Bohr theory can be successfully modified to explain the electronic spectrum of multielectron atoms.
- (D) The orbital angular momentum of an electron in an atom is given as $\frac{nh}{2\pi}$.

Space for Rough Work

2. For the reaction : $A \rightarrow B$. Differential rate law is : $\frac{-d[A]}{dt} = k[A]^{-1}$. If initial concentration of A is A_0 M and concentration at time 't' is A_t M then identify correct option (s)

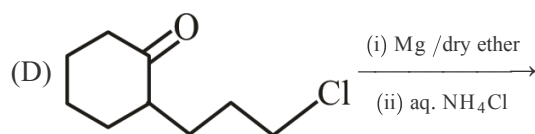
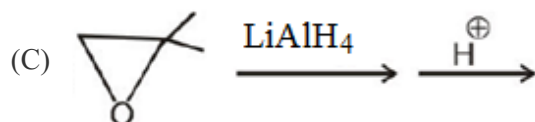
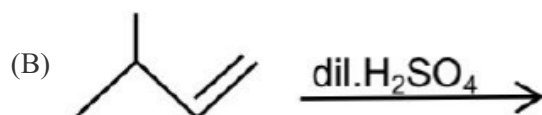
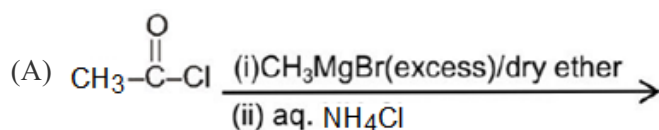
(A) $A_t^2 = A_0^2 - 2kt$

(B) Half life, $t_{1/2} = \frac{3A_0^2}{4K}$

(C) Completion time for reaction : $t_{100\%} = \frac{A_0^2}{2K}$

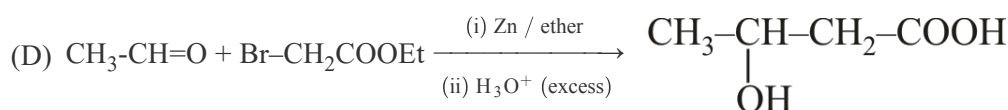
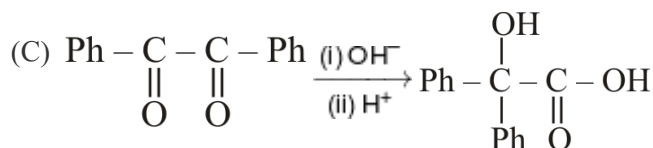
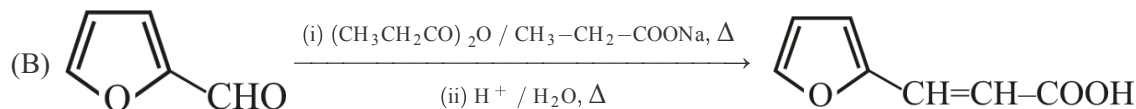
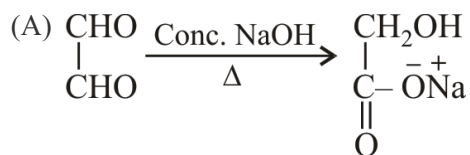
(D) The reaction must be a complex reaction

3. Which of the following reaction will generate 3° alcohol?



Space for Rough Work

4. Select reaction(s) correctly matched with the major product :



5. Which of the following statement(s) regarding the pair of complexes is/are incorrect?

- (A) Both $\text{K}_2[\text{PdCl}_4]$ and $\text{K}_2[\text{Pt}(\text{ox})_3]$ are optically inactive
 (B) Both $\text{K}_2[\text{PdCl}_4]$ and $\text{K}_2[\text{Pt}(\text{ox})_3]$ do not form diastereomers.
 (C) $\text{K}_2[\text{Ni}(\text{CN})_4]$ and $\text{Co}[\text{Hg}(\text{SCN})_4]$ have same hybridization of central atom.
 (D) $\text{K}_2[\text{Ni}(\text{CN})_4]$ and $\text{Co}[\text{Hg}(\text{SCN})_4]$ both can show same type of structural isomerism and are diamagnetic.

6. Select similarity between Br^- and I^- ion :-

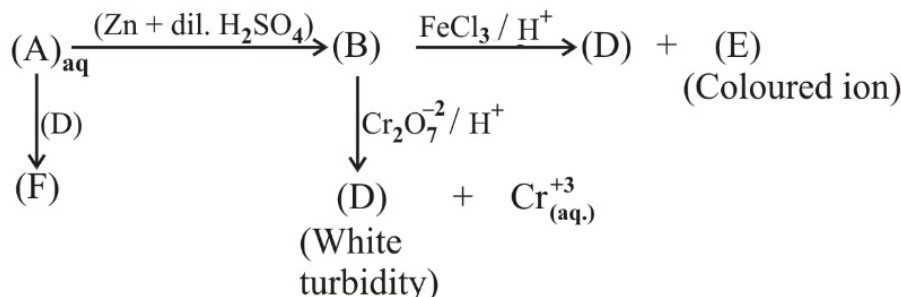
- (A) Both can be oxidised by conc. H_2SO_4 to produce respective halogen
 (B) Both can be oxidised by Cl_2 water to produce respective halogen
 (C) Both can be oxidised by $\text{Cu}_{(\text{aq.})}^{2+}$ to produce respective halogen
 (D) Both are precipitated by AgNO_3 solution

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

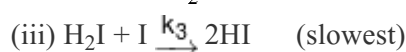
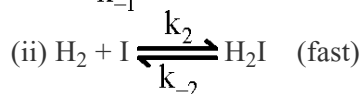


7. Species (A) and (F) are respectively :
- | | |
|--|--|
| (A) $\text{SO}_{3(\text{aq.})}^{-2}$, S | (B) $\text{SO}_{3(\text{aq.})}^{-2}$, $\text{S}_2\text{O}_{3(\text{aq.})}^{-2}$ |
| (C) $\text{S}_2\text{O}_{3(\text{aq.})}^{-2}$, $\text{SO}_{3(\text{aq.})}^{-2}$ | (D) $\text{SO}_{3(\text{aq.})}^{2-}$, $\text{SCN}_{\text{aq.}}$ |
8. When FeCl_3 react with NH_4SCN then :
- | | |
|--------------------------------------|--|
| (A) Blood red complex is formed | (B) Blue coloured complex is formed |
| (C) Brown coloured complex is formed | (D) Green solution of Fe^{2+} |

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a three step mechanism



9. Which is/are incorrect regarding above mechanism ?

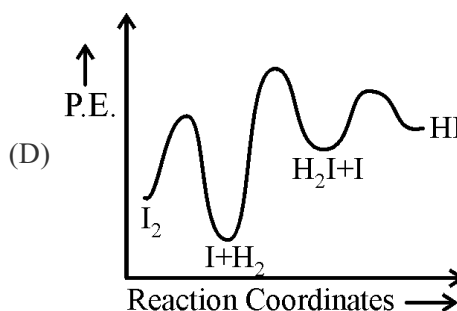
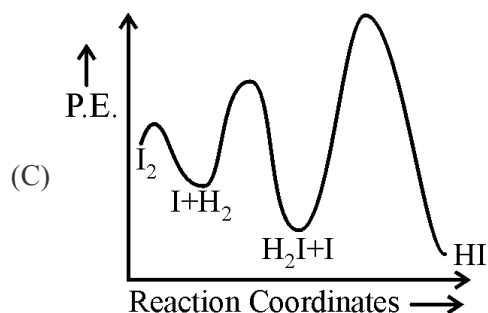
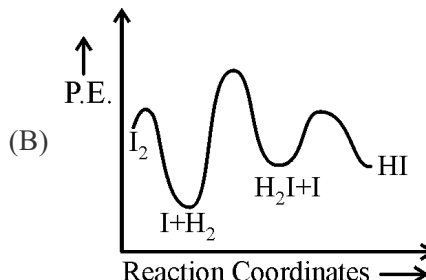
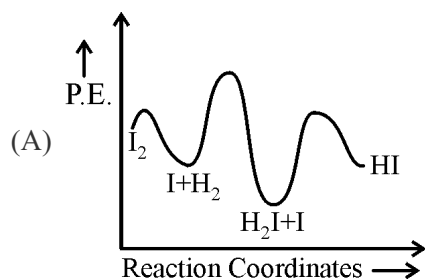
(A) Rate of reaction = $K[\text{H}_2\text{I}][\text{I}]$ where $K = k_3 \cdot \frac{k_2}{k_{-2}} \cdot \left(\frac{k_1}{k_{-1}}\right)$

(B) Rate is decided only by the steps (i) and (ii)

(C) Molecularity of overall reaction is 2

(D) Order of overall reaction is 2.

10. Which of the following qualitative plot of potential energy (PE) v/s reaction coordinates is/are incorrect if $k_1 \gg k_2 \gg k_3$ and reaction is exothermic :



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

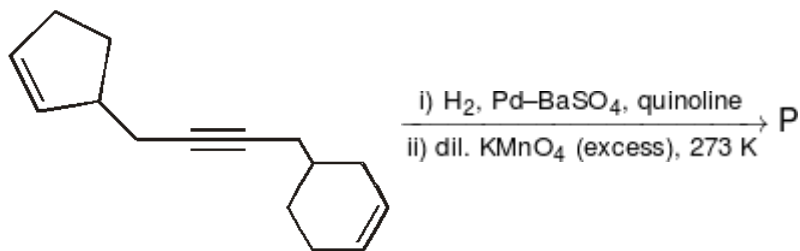
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

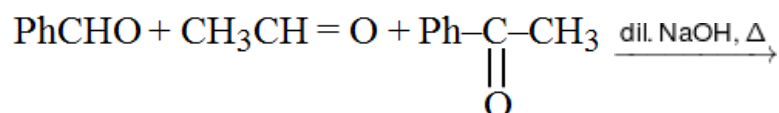
1. A sample of hydrogen atoms contains few atoms in two different excited state n_1 and n_2 only. These atoms make transition to ground state. If the energy of spectral line obtained due to transition of electron from n_2 to n_1 is 2.55 eV then calculate the maximum number of different lines obtained in spectrum.
2. How many of the following statement(s) are true (Assuming all gases involved to behave ideally) ?
 - (i) $\text{H}_2\text{O}(\ell) \longrightarrow \text{H}_2\text{O}(\text{g})$
1 atm, 373K 1 atm, 373K
For this $\Delta H = P\Delta V$ as $\Delta U = 0$
 - (ii) For the ideal gas reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \longrightarrow 2\text{HI}(\text{g})$ occurring in a closed rigid container
 $\Delta H = q_v$ where q_v is the heat absorbed/released at constant volume.
 - (iii) For any irreversible isothermal expansion of an ideal gas $|\Delta S_{\text{system}}| > |\Delta S_{\text{surr}}|$
 - (iv) For an ideal gas $\Delta H = nC_{p,m}\Delta T$ is applicable only under constant pressure condition for temperature change ΔT .
 - (v) The change in enthalpy of the gas is (a) zero if it is expanded reversibly with $T_i = T_f$ and (b) positive if it is expanded reversibly under adiabatic condition with $T_i \neq T_f$.
 - (vi) If the expansion is carried out freely, no change in internal energy, enthalpy and entropy is observed.
 - (vii) $\text{P (black, solid)} \longrightarrow \text{P (white, solid)}$ $\Delta S = \text{positive}$
 - (viii) For mixing equal volumes of two ideal gases at constant temperature and pressure in an isolated container, $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{total}} > 0$

Space for Rough Work

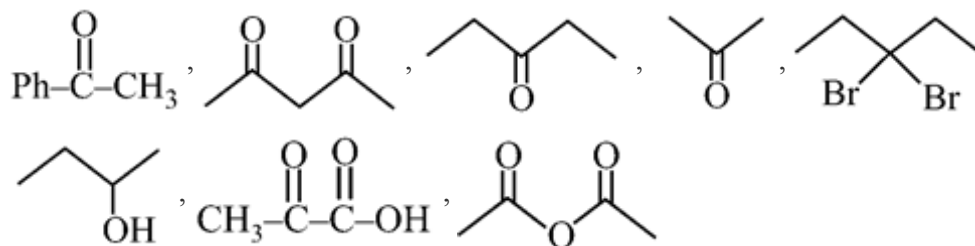
3. Total number of hydroxyl groups present in a molecule of the major product P is ____



4. The total number of all possible Mono Aldols (β -Hydroxy carbonyl compounds) are formed in the following reaction. (Ignore stereoisomers)

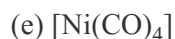
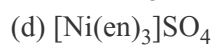
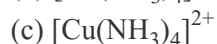
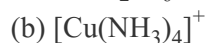
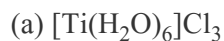


5. How many compounds gives iodoform test with $\text{NaOH} + \text{I}_2$:



Space for Rough Work

6. Select the complexes which will absorb in visible range :



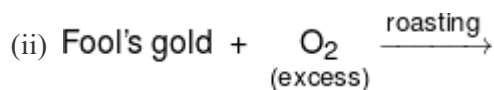
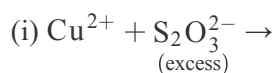
Hence, write total number of such complexes :

7. Identify total number of compounds which liberate 'CH₄' gas on reaction with CH₃MgCl.



8.

Find sum of the oxidation number of metal in products of given reactions.



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	:	+4	If only (all) the correct option(s) is (are) chosen.
<i>Partial Marks</i>	:	+3	If all the four options are correct but ONLY three options are chosen.
<i>Partial Marks</i>	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
<i>Partial Marks</i>	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
<i>Zero Marks</i>	:	0	If none of the options is chosen (i.e. the question is unanswered).
<i>Negative Marks</i>	:	-1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let $f(x) = \cot^{-1} (x^2 + 4x + \alpha^2 - 3\alpha)$ defined on $\mathbb{R} \rightarrow \left(0, \frac{\pi}{2}\right]$, is an onto function then

- (A) $\alpha \in [-1, 4]$ (B) $f'(0) = -4/17$
(C) $f(x)$ is one-one function (D) $f(x)$ is many-one function

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x + 2x^2 \sin \frac{1}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ then

- (A) $f(x)$ is differentiable for all x , but $f'(x)$ is not continuous at $x = 0$
(B) $f'(0) = 1$
(C) $f(x)$ is increasing at $x = 0$
(D) Both $f(x)$ and $f'(x)$ are differentiable for all x .

Space for Rough Work

3. Area bounded by curves $y = \sqrt{|x-1|}$ and $\sqrt{y} = |x-1|$ is

(A) $2 \int_1^2 (\sqrt{x-1} - (x-1)^2) dx$

(B) $2 \int_0^1 (\sqrt{x} - x^2) dx$

(C) $2 \int_1^2 (\sqrt{2-x} - (2-x)^2) dx$

(D) $\frac{2}{3}$

4. Let $f(x) = x \ln\left(1 + \frac{1}{x}\right)$, $0 < x < \infty$, then which of the following is correct

(A) f is strictly increasing

(B) $\lim_{x \rightarrow 0} f(x) = 0$

(C) $\lim_{x \rightarrow \infty} f(x) = 1$

(D) f has exactly one point of extremum

Space for Rough Work

5. Let $f(x) = 2x^3 - 15x^2 + 36x - 23$ and $g(x) = \begin{cases} \max \{f(t), 1 \leq t \leq x\}, & \text{if } 1 \leq x \leq \frac{7}{2} \\ 12 - 2x, & \text{if } \frac{7}{2} < x \leq 6 \end{cases}$. Which of the following is correct about $g(x)$?
- (A) Continuous everywhere (B) Differentiable everywhere
- (C) has greatest value 5 (D) has least value 0
6. Let C be a curve such that normal at any point P on it meets x-axis and y-axis at A and B respectively. If $BP : PA = 1 : 2$ (internally) and curve passes through (0,4), then-
- (A) The curve 'C' passes through $(\sqrt{10}, -6)$
- (B) Equation of tangent at $(4, 4\sqrt{3})$ is $\sqrt{3}x - 2y + 4\sqrt{3} = 0$
- (C) Equation of tangent at $(4, 4\sqrt{3})$ is $2x - \sqrt{3}y + 4 = 0$
- (D) If point P is (8,12), then point A and point B are (24,0) and (0,18) respectively.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 7 and 8

For any geometrical figure possessing a finite area 'A' and a finite perimeter 'P', the ratio $\frac{4\pi A}{P^2}$ is called the isoperimetric quotient (I.Q.)

7. If parallelogram ABCD and the triangle ABC have the same I.Q., then the ratio $\frac{AB + BC}{AC}$ is equal to :
- (A) $1 + 2\sqrt{2}$ (B) $\frac{1}{\sqrt{2} - 1}$ (C) $\frac{4 + \sqrt{2}}{2}$ (D) $1 + \sqrt{2}$
8. A sector of a circle has a fixed radius r and a variable angle θ (in radians). The value of θ for which I.Q. of the sector is maximum is :
- (A) 2 (B) $\frac{\pi}{2}$ (C) 1 (D) An integer

Space for Rough Work

Paragraph for Questions 9 and 10

Suppose that function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the inequality, $\left| \sum_{k=1}^n (3^k \{f(x+ky) - f(x-ky)\}) \right| \leq 1$ for every positive integer n and for all $x, y \in \mathbb{R}$. Also, let $g(x) = ax^3 - (3+b)x^2 + cx$ for $x \in \mathbb{R}$, where a, b, c are such that $\lim_{x \rightarrow 0} \frac{axe^x - b \ln(1+x) + cxe^{-x}}{x^2 \sin x} = 2$

9. If $f(13) = -23$, number of solutions of $g(x) = f(x)$ is more than or equal to :
 (A) 2 (B) 3 (C) 4 (D) 1
10. The value of $\lim_{x \rightarrow 1} \frac{g(x) + 3}{\ln(2-x)}$ is equal to
 (A) 12 (B) 3 (C) 9 (D) 1

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let parabola P passes through the points (0, 1), (2, 0), (0, 2) and (2, 2). If axis of symmetry of the parabola has slope m, then |m| is equal to _____.
2. A ray emanating from the point (0, 6) is incident on the ellipse $25x^2 + 16y^2 = 1600$ at the point P with ordinate 5. After reflection, ray cuts the Y-axis at B. The length of PB/2 is equal to _____.
3. The value of the integral $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$ is equal to _____.
4. Let f(x) be a polynomial with leading coefficient unity and satisfying the equation $(x - 16) f(2x) = 16(x - 1) f(x)$ then $\frac{f(0)}{128}$ is equal to _____.

Space for Rough Work

5. $f : [-1, 1] \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$, $f(x) = \frac{x}{1+x^2}$. If the area bounded by $y = f^{-1}(x)$, X-axis between, $x = \frac{-1}{2}$ to $x = \frac{1}{2}$ is $A - \ln 2$, then the value of A is equal to _____.
6. $f(x)$ is a fifth order polynomial in x with every root of $f(x) = 0$ is real and distinct, then the number of real roots of $f''(x) f(x) - (f'(x))^2 = 0$ is equal to _____.
7. If $\int (x^9 + x^6 + x^3) (2x^6 + 3x^3 + 6)^{1/3} dx = a(2x^9 + 3x^6 + 6x^3)^{4/3} + C$, then the value of $48a$ is equal to _____.
(Where C is constant of integration)
8. If $\int \frac{\sec x (2 + \sec x)}{(1 + 2 \sec x)^2} dx$ is $\left(\frac{\cos x + 2}{\lambda \sin x}\right)^p$, then $\lambda + p$ is equal to _____.

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work